MICROSOFT
Software Engineer

Role Practice Assessment

20 original prompts with space to plan your responses

Scenario lens: Fictional product work related to cloud and collaboration. This pack uses general skills expected in software engineer work. It is independently written, is not an official hiring test, and is not affiliated with Microsoft.

20 prompts	75-90 minutes	100-point review guide
Five skill areas	Show your reasoning	Use the rubric at the end

How to use this pack

Answer each prompt in the space provided. State assumptions, show calculations or pseudocode where useful, and explain trade-offs. Use a separate sheet if you need more room. The focus labels are a self-review guide, not a company scoring key.

Suggested review rubric

Technical correctness 35 points · reasoning and trade-offs 25 · validation and edge cases 20 · clarity and communication 20

Name: _____ Date: _____

Algorithms

Fictional cloud and collaboration scenario · answer all four prompts in this section.

01

Given a stream of events, return the k most frequent event types. Describe an efficient approach and explain how you handle ties.

Focus: Complexity, data structures, deterministic tie handling

02

Detect whether a dependency graph contains a cycle, then return one cycle as evidence. State the graph representation you would use.

Focus: Graph traversal, cycle reconstruction, disconnected components

03

For a sorted list of time intervals, merge all overlapping intervals. Walk through adjacent, identical, empty, and very large inputs.

Focus: Sorting, boundary conditions, time and space complexity

04

Design a function that finds the longest substring with no repeated characters. Explain how its window moves and why it is linear time.

Focus: Sliding window invariant, Unicode considerations, complexity

Services and APIs

Fictional cloud and collaboration scenario · answer all four prompts in this section.

05 Sketch an API for saving a user's cloud and collaboration preferences. Include request/response examples, validation, and an approach to versioning.

Focus: Resource design, validation, compatibility

06 A client retries a purchase or save request after a timeout. Design an idempotency strategy that prevents duplicate writes.

Focus: Idempotency keys, persistence, retry semantics

07 A popular cloud and collaboration resource receives sudden traffic spikes. Propose a cache design and explain freshness and invalidation trade-offs.

Focus: Cache keys, TTLs, invalidation, stampede control

08 Design a paginated endpoint for a dataset that changes while users browse. Compare offset and cursor pagination.

Focus: Stable ordering, cursors, consistency

Data and distributed systems

Fictional cloud and collaboration scenario · answer all four prompts in this section.

09 Model users, projects, and memberships for a collaboration feature in cloud and collaboration. Explain keys, constraints, and indexes.

Focus: Relational modeling, constraints, query patterns

10 A background queue delivers the same task more than once. Explain how the worker can safely process it.

Focus: At-least-once delivery, deduplication, transaction boundaries

11 Two regions accept profile updates at nearly the same time. Define a conflict-resolution policy and its user-visible consequences.

Focus: Consistency model, ordering, conflict handling

12 A database table has grown to hundreds of millions of rows. Describe how you would investigate and improve a slow query.

Focus: Query plan, indexes, measurement, safe rollout

Reliability and security

Fictional cloud and collaboration scenario · answer all four prompts in this section.

13 A cloud and collaboration service has rising p95 latency while average latency is unchanged. Give a step-by-step investigation plan.

Focus: Tail latency, traces, saturation, hypotheses

14 Design a per-user rate limiter for a public API. Explain behavior across multiple application servers.

Focus: Token bucket or equivalent, shared state, limits

15 How should a service handle a dependency that is timing out? Describe timeouts, retries, circuit breaking, and fallbacks.

Focus: Deadlines, bounded retries, cascading-failure prevention

16 Review a feature that stores user-generated links or text. Identify security risks and safeguards before launch.

Focus: Authorization, injection/XSS, validation, privacy

Testing and engineering judgment

Fictional cloud and collaboration scenario · answer all four prompts in this section.

17 Write a test plan for an API that changes a user's cloud and collaboration notification settings. Include unit, integration, and end-to-end cases.

Focus: Behavior coverage, persistence, permissions, failure paths

18 A refactor passes unit tests but causes a production regression. Explain how you would narrow the cause and reduce rollout risk.

Focus: Reproduction, observability, rollback, progressive delivery

19 Choose service-level indicators for a feature that returns personalized results. Define useful latency and correctness measures.

Focus: User-centered SLIs, percentiles, measurable correctness

20 Describe a code review comment that improves maintainability without blocking a safe delivery. What evidence would you request?

Focus: Specific feedback, risk, tests, trade-offs

Self-review and submission checklist

Before sharing your work, use this independent review guide. It is not an official company rubric.

Area	Points	Check your response
Technical correctness	35	Does the solution answer the prompt and use sound methods?
Reasoning and trade-offs	25	Are assumptions and alternatives explained?
Validation and edge cases	20	Did you check data quality, boundaries, and failure cases?
Clarity and communication	20	Can a teammate follow the result and proposed next step?
Total	**100**	Use feedback to choose what to practice next.

My strongest area

One skill I will practice next